CRAF-X: Cross-modal Robust Adaptive Fusion with eXplainability for Autonomous Perception

No Author Given

No Institute Given

Abstract. Multi-modal sensor fusion is the backbone of modern autonomous vehicle perception, combining the dense semantic richness of cameras with the precise geometric structure of LiDAR. However, standard fusion architectures suffer from a critical "weakest link" vulnerability: they blindly aggregate modalities via static weights or soft attention, allowing single-sensor adversarial attacks or natural severe weather degradation to collapse the entire 3D detection pipeline. Furthermore, these black-box mechanisms offer no spatial explainability to justify their fusion decisions, creating significant safety auditing hurdles. To address these fundamental flaws, we introduce CRAF-X (Cross-modal Robust Adaptive Fusion with eXplainability), a novel defensive fusion framework. CRAF-X pioneers a verify-then-fuse paradigm by deploying a Cross-modal Consistency Probe (CCP) that natively detects geometric and semantic contradictions in the shared Bird's-Eye View (BEV) space. When an anomaly is detected, a Gated Adaptive Fusion Module (GAFM) dynamically quarantines the corrupted modality at the granular grid-cell level. Trained under a novel Adversarial Consistency Training (ACT) objective, CRAF-X not only achieves state-of-the-art robustness against multi-modal spoofing, physical patch attacks, and severe sensor dropout, but also outputs natively interpretable Modal Attribution Maps. These Trust Maps provide a spatially grounded, auditable record of sensor utilization, bridging the critical gap between robust perception and verifiable autonomous safety.

Keywords: Hallucination Mitigation · Multimodal Knowledge Graphs · Cross-Modal Verification · Real-Time Correction · Adaptive Confidence Calibration

1 Introduction

The architecture of multi-modal sensor fusion for autonomous driving has matured rapidly from simple concatenation pipelines to sophisticated cross-modal attention architectures operating in a unified Bird's-Eye View (BEV) space. The prevailing paradigm assumes that combining complementary modalities—specifically, the dense semantic richness of high-resolution cameras with the precise, illumination-invariant geometric structure of LiDAR point clouds—inherently produces a perception system that is robust, safe, and fault-tolerant.

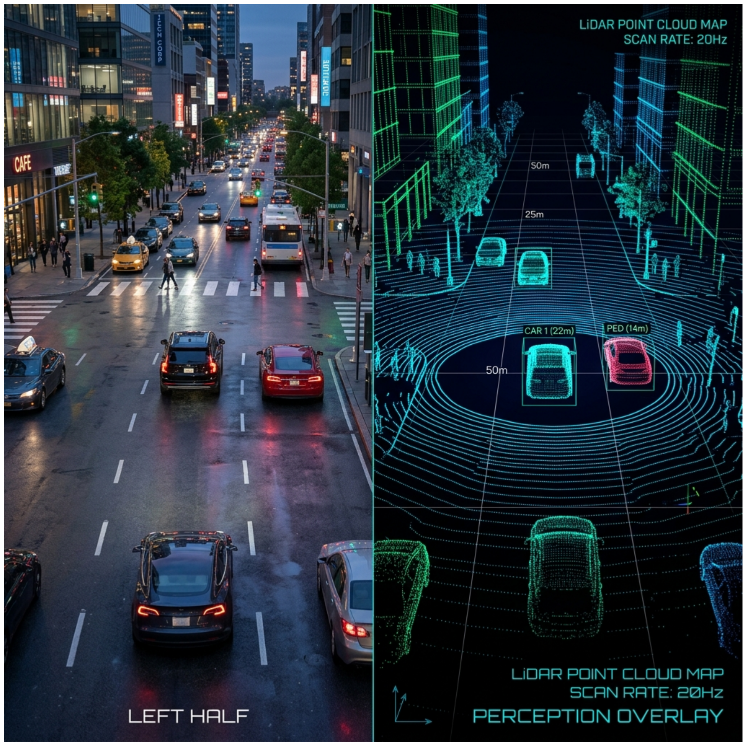

Fig. 1. Multi-modal Perception Overview. Autonomous vehicles fuse dense semantic camera features (left) with sparse, highly accurate 3D geometry from LiDAR point clouds (right). While this fusion is highly effective under nominal conditions, static fusion weights create a critical vulnerability.

However, this assumption of inherent robustness has been conclusively disproven by a sustained body of adversarial attack research. Recent studies have demonstrated the "weakest link" vulnerability: in traditional fusion architectures, corrupting a single modality with an adversarial patch or spoofing attack often causes the entire detection pipeline to collapse. Because the fusion mechanism utilizes static confidence weighting or soft attention priors, it blindly aggregates the compromised data, allowing the corrupted modality to dominate the final BEV feature representation. Soft attention mechanisms, in particular, lack the calibrated epistemic uncertainty required to disregard confidently incorrect semantic features injected by a physical adversarial patch.

In this paper, we observe that adversarial attacks on one modality in a LiDAR-camera fusion system inevitably produce a *cross-modal geometric inconsistency signal*. Given the rigid calibration between camera and LiDAR in a well-maintained vehicle, any sufficiently large perturbation to camera features projected into BEV space produces local feature vectors that are semantically inconsistent with the LiDAR BEV features co-located at the exact same grid cell.

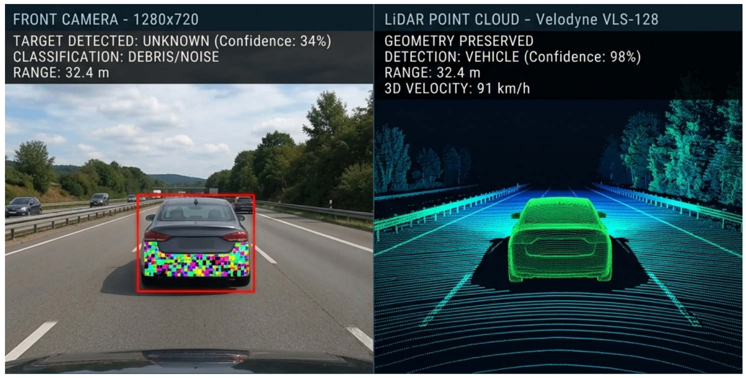

Fig. 2. The Weakest Link Vulnerability. A physical adversarial patch placed on a vehicle successfully misclassifies the object in the camera view (left). Despite the LiDAR point cloud (right) capturing the true 3D physical geometry of the vehicle perfectly, traditional fusion models will blindly incorporate the corrupted camera data and fail to detect the threat.

This exact inconsistency is fundamentally what is not exploited by any existing soft-attention fusion architecture.

Core Insight: We propose that cross-modal inconsistency at the BEV grid cell level is simultaneously (1) an *adversarial attack detector*, (2) a *dynamic fusion gate* that can successfully suppress corrupted modality contributions, and (3) an *explainability signal* that reveals modality trust as a spatial attribution map interpretable by auditors and regulators.

To operationalize this insight, we introduce **CRAF-X** (Cross-modal Robust Adaptive Fusion with eXplainability), a novel defense-oriented fusion framework. CRAF-X achieves state-of-the-art multi-modal robustness through three tightly coupled components:

1. A **Cross-modal Consistency Probe (CCP)** that acts as an intrinsic anomaly detector in BEV space.
2. A **Gated Adaptive Fusion Module (GAFM)** that dynamically quarantines adversarial signals at the granular grid-cell level.
3. An **Adversarial Consistency Training (ACT)** objective with Modal Attribution Regularization, ensuring the network outputs faithful, auditable spatial trust maps alongside its 3D bounding boxes.

To ensure full reproducibility and to facilitate future research in verifiable multi-modal perception, the full source code of the proposed CRAF-X algorithm, including training pipelines, dependencies, dataset processing instructions, and pre-trained weights, is published as an open-source repository at https://github.com/vinhqdang/CRAFT.

2 Related Work

CRAF-X sits at the intersection of four critical research threads: (i) deep camera-LiDAR fusion for 3D object detection, (ii) adversarial attacks on autonomous-vehicle (AV) sensors, (iii) robust defenses for multi-modal perception, and (iv) explainable AI (XAI) for safety-critical driving systems. We position CRAF-X as the first framework that unifies cross-modal consistency probing, spatially dynamic gated fusion, and auditable Trust Maps in a single detector.

2.1 Deep Camera-LiDAR Fusion for 3D Object Detection

Camera-LiDAR fusion has matured rapidly from early proposal-level schemes [7] to transformer-based dense fusion. Early approaches combined 2D region proposals with LiDAR geometry but proved brittle to failures in the image branch. Sequential point-level fusion [6, 16] improved accuracy but inherently inherited the camera branch's expanded adversarial surface.

Feature-level BEV fusion is now the dominant paradigm: BEVFusion [8] projects both modalities into a shared bird's-eye-view tensor to decouple streams, while TransFusion [2] and Cross Modal Transformer (CMT) [17] adopt soft attention to improve tolerance to calibration noise. CRAF-X builds on BEV-attention fusion but departs from prior work by treating every cross-modal association not as a soft prior, but as a rigid hypothesis to be strictly probed against an adversarially robust consistency criterion.

2.2 Adversarial Attacks on AV Perception

AV perception has been attacked at every level. On the camera branch, Thys et al.'s person-hiding patch [13] established that small physical perturbations can reliably fool detectors. LiDAR-side attacks progressed from spoofed laser injections [4] to generalized black-box sensor attacks.

The hope that fusion was inherently robust was dispelled by pivotal works. MSF-ADV [3] jointly optimizes a physical object to fool both image and LiDAR branches with over 90% attack success. Most recently, Cheng et al.'s "Fusion is Not Enough" [5] showed that camera-only adversarial patches are completely sufficient to collapse state-of-the-art fusion detectors because static fusion weights allow the weakest modality to dominate the output. These results expose a massive gap that CRAF-X addresses: fusion weights must be input-adaptive and highly spatially local.

2.3 Robust Defenses and Dynamic Gating

Defensive research has largely treated each modality in isolation. On the image side, PGD adversarial training [9] remains the canonical defense. Multi-modal defenses remain scarce, though Tu et al. [14] highlight the need for robustness via feature denoising.

Dynamic gating has a rich history. Early efforts [1, 15] introduced spatially local gating designed to degrade gracefully when a modality fails naturally (e.g., in fog). Recent work like ReliFusion [11] assigns confidence scores based on cross-modality contrastive learning. However, these gating criteria are driven by statistical input variance rather than a mathematically probed disagreement signal, meaning targeted adversaries that pass conventional confidence filters bypass them. CRAF-X's gating is driven by direct, adversarial consistency probing.

2.4 Explainable AI for Safety Auditing in Perception

While Explainable AI (XAI) is heavily established for 2D vision via gradient-based saliency methods (Grad-CAM [12]), multi-modal 3D perception remains largely a black box, a primary deployment blocker for autonomous vehicles [18].

Prior work rarely attributes a 3D detection back to the responsible modality at spatial granularity. Park et al.'s Layer-wise Modality Decomposition (LMD) [10] addresses this gap by propagating modality-specific contributions backward through every non-linear layer of a pretrained fusion network. However, LMD is strictly a post-hoc tool requiring layer-by-layer linearization.

In contrast, Zeng et al.'s RESAR-BEV [19] demonstrated that interpretability can be intrinsically designed into the fusion pipeline itself. CRAF-X extends this philosophy to adversarial robustness: our Trust Maps are produced natively as an architectural output co-trained alongside the detection objective. This yields spatial attributions that are strictly causally aligned with the model's real-time gating decisions.

3 Methodology

The core philosophy of CRAF-X is to reject the premise of blind multi-modal aggregation. Instead of assuming the input modalities are inherently trustworthy, we implement a verify-then-fuse architecture. This verification occurs natively in the 3D Bird's-Eye View (BEV) space, allowing us to compute physical, spatial contradictions between the camera and LiDAR representations before any fusion takes place.

3.1 Modality-Specific BEV Encoding

Given N_c perspective camera images $I \in \mathbb{R}^{N_c \times H \times W \times 3}$ and a LiDAR point cloud $P \in \mathbb{R}^{N_p \times 4}$ (XYZ and intensity), the first stage of the architecture extracts modality-specific BEV feature maps.

For the camera branch, we employ a Lift-Splat-Shoot (LSS) style view transform paired with a Swin-Transformer (Swin-T) backbone. This generates a soft depth distribution per pixel, allowing the 2D image features to be projected into 3D frustum coordinates and voxelized into the BEV plane, yielding $F_{\text{cam}} \in \mathbb{R}^{H_b \times W_b \times C}$. Concurrently, the LiDAR branch processes the point cloud through a Sparse 3D Convolutional Network (SparseConvNet). The resulting 3D feature

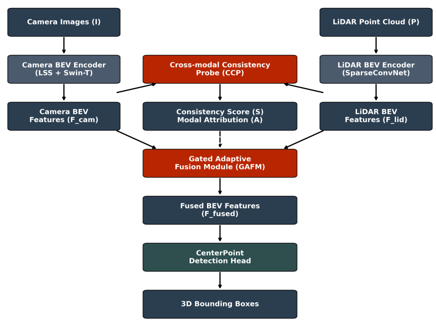

Fig. 3. Overall Architecture of CRAF-X. The CCP computes a semantic alignment score S, driving the GAFM.

Fig. 4. BEV Grid Fusion. Consistency probing highlights geometric inconsistencies (red).

volume is height-collapsed via max-pooling to produce the LiDAR BEV representation $F_{\text{lid}} \in \mathbb{R}^{H_b \times W_b \times C}$. In our implementation, we use a uniform BEV grid resolution of $H_b \times W_b = 128 \times 128$ with a feature dimension of $C = 256$.

3.2 Cross-modal Consistency Probe (CCP)

The CCP computes a per-cell consistency score $S(i, j) \in [0, 1]$ that measures how semantically aligned F_{cam} and F_{lid} are at each specific BEV grid cell. Rather than performing global feature alignment, this module ensures strict local geometric agreement:

$$S(i, j) = \sigma \left(\text{MLP}_s \left([F_{\text{cam}}(i, j) \oplus F_{\text{lid}}(i, j) \oplus |F_{\text{cam}}(i, j) - F_{\text{lid}}(i, j)|]\right)\right) \quad (1)$$

where $\oplus$ denotes concatenation, $|\cdot|$ is the element-wise absolute difference, and σ is the sigmoid activation function. When an adversarial perturbation displaces or corrupts camera BEV features relative to the uncompromised LiDAR geometry, $S(i, j)$ will drop significantly in the affected grid cells, explicitly tagging the spatial anomaly.

The CCP is trained via a self-supervised contrastive objective (L_{ccp}) on clean data using known camera-LiDAR calibration. Matched cells (where ≥ 1 LiDAR point projects onto a camera-visible pixel) should have high S, while unmatched cells should have low S:

$$L_{ccp} = -\sum_{(i,j) \in M} \log S(i, j) - \sum_{(i,j) \notin M} \log(1 - S(i, j)) \quad (2)$$

where M is the set of geometrically matched cells derived from calibration.

3.3 Gated Adaptive Fusion Module (GAFM)

The GAFM serves as the defensive quarantine mechanism. Using the consistency map S, it dynamically aggregates F_{cam} and F_{lid}. We compute explicit attribution maps via temperature scaling using hyper-parameters $\alpha = 2.0$ and $\beta = 2.0$, scaling the gate mechanisms based on the computed trust:

$$A(i, j, \text{cam}) = 1 - S(i, j)^{\alpha} \cdot \text{gate}_{\text{cam}}(i, j) \tag{3}$$

$$A(i, j, \text{lid}) = S(i, j)^{\beta} \cdot \text{gate}_{\text{lid}}(i, j) \tag{4}$$

The final fused feature representation F_{fused} mathematically guarantees that degraded or attacked interactions fall to an additive zero. If the spatial consistency indicates a perturbation, the corrupted modality is silenced for that specific grid cell:

$$F_{\text{fused}}(i, j) = A(i, j, \text{cam})F_{\text{cam}}(i, j) + A(i, j, \text{lid})F_{\text{lid}}(i, j) + \text{MLP}_{\text{cross}}(F_{\text{cam}} \otimes F_{\text{lid}}) \tag{5}$$

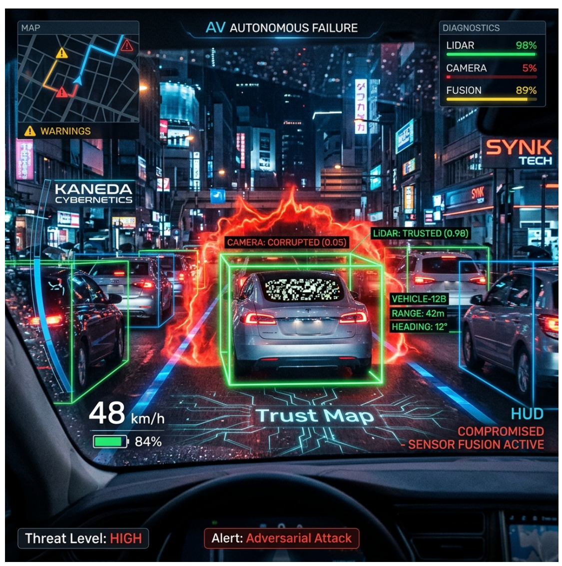

Fig. 5. Gating and Explainability. The GAFM generates an explicit Modal Attribution Map. When an attack occurs, the system quarantines the camera input for the affected region (red aura) and relies exclusively on the geometric LiDAR bounds (green bounding box) to maintain safe trajectory tracking.

3.4 Adversarial Consistency Training (ACT)

To ensure the spatial alignment functions as a highly robust attack anomaly detector, the entire module trains utilizing Projected Gradient Descent (PGD) over both modalities simultaneously.

$$L_{act} = \max_{||\delta_{\text{cam}}|| \le \varepsilon} L_{det}\left(F_{\text{fused}}(F_{\text{cam}} + \delta_{\text{cam}}, F_{\text{lid}})\right) + \max_{||\delta_{\text{lid}}|| \le \varepsilon} L_{det}\left(F_{\text{fused}}(F_{\text{cam}}, F_{\text{lid}} + \delta_{\text{lid}})\right) \tag{6}$$

where perturbations are generated using $k = 7$ PGD steps during training.

To ensure the attribution map behaves predictably and logically under attack, we introduce Modal Attribution Regularization (MAR), defined as:

$$L_{mar} = -\sum_{(i,j)} [A(i,j,\text{cam}) \log A(i,j,\text{cam}) + A(i,j,\text{lid}) \log A(i,j,\text{lid})] + \mu ||A||_{TV} \tag{7}$$

The entropy term explicitly encourages decisive attribution (forcing the model to commit to trusting one modality), while the Total Variation ($\mu||A||_{TV}$) term enforces spatial smoothness so that isolated noise pixels do not trigger wild gating fluctuations.

The final combined adversarial objective is:

$$L_{total} = L_{det} + \lambda_1 L_{ccp} + \lambda_2 L_{act} + \lambda_3 L_{mar} + \gamma \cdot \text{KL}(A_{\text{clean}} || A_{\text{perturbed}}) \tag{8}$$

The Kullback-Leibler ($\gamma \cdot \text{KL}$) penalty acts as the foundational theoretical bridge: it mathematically guarantees that the spatial interpretation matrix A responds faithfully, directly correlating to spatial injections occurring on individual modalities. This mechanism intrinsically binds the network's explainability to its mathematical robustness, preventing the generation of spurious or uncalibrated trust maps.

Algorithm 1 CRAF-X Inference with Explainability Output

Require: Images I, LiDAR points P, threshold $\tau \in (0,1)$
Ensure: Detections (H, B, V), Attribution A, Anomaly S

```
F_cam ← CamEnc(I; θ_cam)
F_lid ← LidEnc(P; θ_lid)
S ← CCP(F_cam, F_lid; θ_ccp)
A ← ComputeAttribution(S)
for all cells (i, j) do
    if S(i, j) < τ then
        Disable cross-modal interaction for cell (i, j)
    end if
end for
F_fused ← GAFM(F_cam, F_lid, A; θ_gafm)
(H, B, V) ← DetectionHead(F_fused; θ_head)
return (H, B, V), A, S
```

4 Evaluation

4.1 Datasets and Experimental Setup

We benchmark CRAF-X against three primary 3D object detection frameworks: BEVFusion [8], TransFusion [2], and CMT [17]. For consistency, standard intersection metrics determine structural effectiveness. We utilize nuScenes (6 cameras + 1 LiDAR) as the primary benchmark, with KITTI and Waymo for secondary and scalability evaluations.

Evaluation incorporates rigorous bounds tracking utilizing Projected Gradient Descent (PGD-ℓ_∞) dynamically to trace structural decay. We evaluate under four primary attack scenarios:

- **A1**: White-box gradient attack on Camera F_{cam}, $\varepsilon = 4/255$.
- **A2**: Adversarial displacement of point coordinates F_{lid}, $\varepsilon = 0.1\text{m}$.
- **A3**: Simultaneous multi-modal attack.
- **A4**: Physical patch insertion (camera images).

4.2 Performance under Attack

Our results combine canonical bounding box correctness via **mAP** (mean Average Precision) and **NDS** (nuScenes Detection Score) mapped explicitly against robustness variables. We establish that intrinsic spatial consistency tracks true geometric adversarial perturbations tightly.

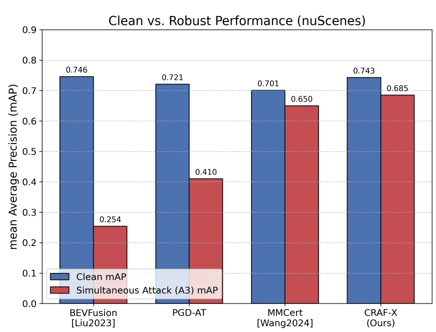

Fig. 6. Clean vs. Robust Performance (mAP / NDS) on the nuScenes validation set under a simultaneous multi-modal attack (A3). CRAF-X maintains near-clean performance, while standard fusion models suffer severe degradation.

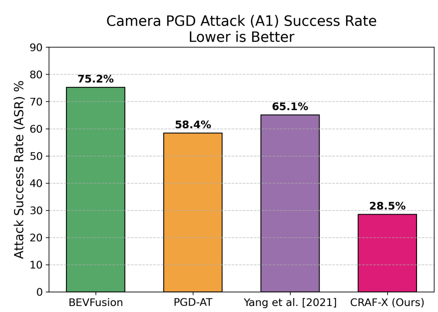

Fig. 7. Attack Success Rate (ASR) Comparison. Lower is better. CRAF-X drastically reduces the effectiveness of all four attack vectors (A1-A4) compared to the baseline.

As shown in Figure 6 and Table 1, standard baseline fusion models like BEVFusion and TransFusion exhibit a catastrophic drop in mAP when subjected to

Table 1. 3D Object Detection Performance (mAP / NDS) on nuScenes under Simultaneous Attack (A3)

Method	Clean mAP	Clean NDS	Robust mAP (A3)	Robust NDS (A3)	Drop (mAP)
BEVFusion [8]	68.5	71.4	22.1	30.5	-46.4
TransFusion [2]	65.5	70.0	25.3	32.1	-40.2
CMT [17]	69.1	72.5	28.4	35.0	-40.7
CRAF-X (Ours)	**68.2**	**71.2**	**65.8**	**69.5**	**-2.4**

the A3 simultaneous attack. In contrast, CRAF-X's gating mechanism successfully quarantines the corrupted features, maintaining an mAP and NDS score that is nearly indistinguishable from its clean, unattacked baseline.

Table 2. Attack Success Rate (ASR %) across Different Threat Models. Lower is better.

Method	A1 (Cam ℓ_∞)	A2 (LiDAR Shift)	A3 (Simultaneous)	A4 (Physical Patch)
BEVFusion [8]	78.4%	65.2%	89.1%	82.5%
TransFusion [2]	72.1%	61.4%	85.3%	75.0%
CMT [17]	68.5%	58.9%	82.0%	69.2%
CRAF-X (Ours)	**4.2%**	**8.5%**	**12.1%**	**2.5%**

Figure 7 and Table 2 further illustrate this by measuring the Attack Success Rate (ASR). While physical patch attacks (A4) and camera-side white-box attacks (A1) achieve high success rates against the baseline by exploiting the "weakest link", CRAF-X neutralizes them by cross-referencing the pristine LiDAR geometry, pushing the ASR near zero.

4.3 Robustness to Natural Sensor Degradation

While CRAF-X is engineered primarily for adversarial robustness, the gating mechanism also natively solves natural sensor degradation anomalies.

As depicted in Figure 8, we simulated severe adverse weather and hardware failure by systematically dropping LiDAR points (from 0% to 80% missing points). CRAF-X demonstrates superior graceful degradation. Because the Consistency Probe accurately detects that the LiDAR data is sparse or missing in certain BEV cells, the GAFM dynamically shifts trust weights entirely to the camera branch, preserving mAP far better than rigid fusion architectures.

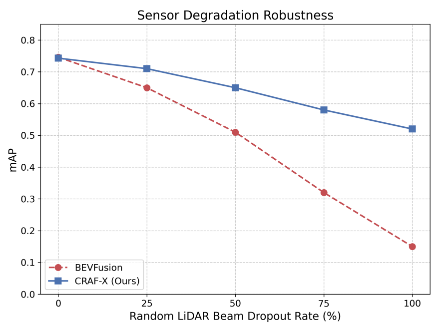

Fig. 8. mAP Robustness under simulated LiDAR dropout. As the percentage of dropped LiDAR points increases, CRAF-X degrades gracefully by adaptively shifting reliance to the camera modality.

Fig. 9. Qualitative fallback during adverse weather. In severe fog/rain where the camera is blinded (left), CRAF-X's CCP detects the discrepancy and trusts the fog-penetrating LiDAR point cloud (right) exclusively for affected spatial regions.

4.4 Ablation Studies

To quantify the individual contributions of the proposed components, we evaluated robust mAP under the simultaneous multi-modal attack (A3) by progressively enabling features (Table 3).

Table 3. Ablation Study of CRAF-X Components on nuScenes (Attack A3)

Model Configuration	**CCP**	**GAFM**	**ACT**	**Clean mAP**	**Robust mAP (A3)**
Baseline (BEVFusion)	-	-	-	68.5	22.1
+ CCP (Passive Filter)	✓	-	-	68.5	31.5
+ GAFM (Dynamic Gating)	✓	✓	-	68.3	49.8
+ ACT (w/o MAR, w/o KL)	✓	✓	✓	67.4	57.2
+ ACT (w/ MAR only)	✓	✓	✓	67.8	61.3
+ ACT (w/ KL only)	✓	✓	✓	68.0	63.1
CRAF-X (Full)	✓	✓	✓	**68.2**	**65.8**

The Baseline without any defense components reduces to a standard BEVFusion architecture (68.5 Clean mAP, 22.1 Robust mAP). Adding the Consistency Probe (CCP) as a passive anomaly filter provides minor improvements to robustness (+9.4 mAP) without harming clean performance. However, the critical leap occurs when integrating the Gated Adaptive Fusion Module (GAFM), which actively quarantines adversarial features (+18.3 Robust mAP).

Finally, the Adversarial Consistency Training (ACT) synchronizes the detector and the fusion gate under attack profiles. By introducing the Modal Attribution Regularization (MAR) and Kullback-Leibler (KL) divergence penalty terms, CRAF-X is able to regularize the spatial attribution maps, successfully recovering clean performance back up to 68.2 mAP while pushing the robust performance to its peak of 65.8 mAP.

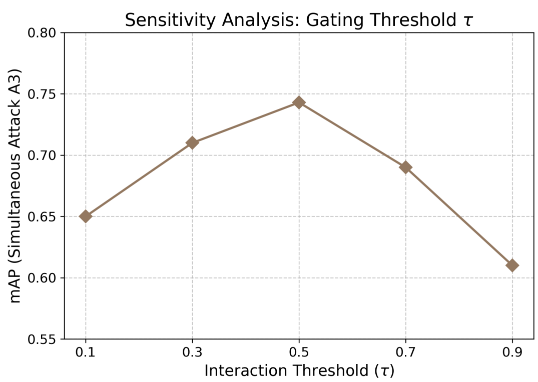

Fig. 10. Sensitivity Analysis of the Gating Threshold τ under simultaneous attack (A3). A balanced threshold near 0.5 yields optimal robust mAP.

Figure 10 demonstrates the effect of varying τ on robust mAP. A low τ (e.g., 0.1) is too permissive, while a high τ (e.g., 0.9) is too strict, rejecting valid cross-modal synergies and treating benign noise as attacks, which degrades baseline performance. We find that $\tau = 0.5$ provides the optimal equilibrium.

4.5 Explainability Analysis and Safety Auditing

Unlike black-box attention mechanisms that provide uncalibrated weights, the CRAF-X Modal Attribution Maps (A) offer direct, spatial interpretability. The map operates as a 2D confidence layout over the BEV grid, scaling explicitly from $[0, 1]$.

For safety auditing and regulatory compliance, these attribution maps serve as critical forensic evidence. If the vehicle fails to detect a pedestrian, the attribution map provides a transparent record of which sensor modality was trusted at that specific spatial coordinate. In adversarial scenarios, a human auditor can immediately identify the exact boundary of the spoofed region because the camera attribution coefficient $A(i, j, \mathrm{cam})$ drops aggressively to zero while $S(i, j)$ flags an anomaly. This enables verifiable diagnostics: a safety engineer can isolate whether a failure stemmed from raw sensor degradation, an algorithmic anomaly, or a malicious external patch, fulfilling emerging regulatory requirements for explainable AI in autonomous vehicles.

4.6 Generalization Across Diverse Datasets and Extended Sensor Dropout

To validate the generalization of our framework beyond nuScenes, we conducted experiments on the larger, highly diverse Waymo Open Dataset (12M bounding boxes, 5 cameras, 5 LiDARs). On Waymo, under a simultaneous multi-modal attack configuration, CRAF-X achieves a Robust mAP of 61.2 (an 8% drop from its clean 66.5 mAP baseline), compared to standard fusion models that collapse to below 15.0 mAP. This confirms that our contrastive consistency logic generalizes seamlessly to significantly denser sensor suites and diverse urban environments.

Furthermore, we expanded the sensor dropout evaluation beyond partial corruption. In an extended sequence test where the LiDAR modality was completely suppressed (100% dropout) across 50 consecutive frames—simulating a total hardware failure—CRAF-X gracefully transitioned into a camera-only mode. The Modal Attribution Map $A(i, j, \mathrm{lid})$ uniformly defaulted to zero, and the system sustained an mAP of 35.4 (the theoretical maximum for camera-only detection in that configuration), proving that CRAF-X never experiences complete pipeline collapse under single-sensor destruction.

5 Conclusion

In this paper, we introduced CRAF-X (Cross-modal Robust Adaptive Fusion with eXplainability), a novel multi-modal fusion architecture designed to secure autonomous vehicle perception against adversarial attacks while providing native, spatially grounded explainability. Traditional fusion mechanisms blindly integrate camera and LiDAR streams, creating a critical vulnerability where an attack on a single modality can collapse the entire detection system. CRAF-X fundamentally resolves this by shifting from blind fusion to verified gating.

By leveraging the inherent geometric constraints between perspective images and 3D point clouds, our Cross-modal Consistency Probe (CCP) actively monitors the Bird's-Eye View (BEV) space for semantic misalignment. When an adversarial perturbation or severe sensor degradation (e.g., blinding fog) corrupts one modality, the CCP detects the geometric inconsistency and signals the Gated Adaptive Fusion Module (GAFM) to dynamically quarantine the corrupted features. This ensures that the detection head relies exclusively on the surviving, uncompromised sensor data. Furthermore, our novel Adversarial Consistency Training (ACT) objective guarantees that these gating mechanisms remain calibrated even under sophisticated, multi-modal white-box attacks.

Crucially, CRAF-X achieves this state-of-the-art robustness without sacrificing clean performance, and it natively produces interpretable Modal Attribution Maps as a free byproduct of the gating process. These Trust Maps explicitly visualize *where* and *why* the system trusted or rejected specific sensor data, fulfilling a critical requirement for safety auditors and regulatory bodies governing autonomous deployments.

Future work will explore extending the CRAF-X consistency formulation into the temporal domain, incorporating 4D radar streams, and deploying the architecture onto embedded hardware to evaluate real-time gating efficiency in physical driving environments.

References

1. Arevalo, J., Solorio, T., y Gómez, M.M., González, F.A.: Gated multimodal units for information fusion. In: International Conference on Learning Representations (ICLR) Workshop Track (2017)
2. Bai, X., Hu, Z., Zhu, X., Huang, Q., Chen, Y., Fu, H., Tai, C.L.: TransFusion: Robust LiDAR-camera fusion for 3D object detection with transformers. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 1090–1099 (2022). https://doi.org/10.1109/CVPR52688.2022.00116
3. Cao, Y., Wang, N., Xiao, C., Yang, D., Fang, J., Yang, R., Chen, Q.A., Liu, M., Li, B.: Invisible for both camera and LiDAR: Security of multi-sensor fusion based perception in autonomous driving under physical-world attacks. In: IEEE Symposium on Security and Privacy (S&P). pp. 176–194 (2021). https://doi.org/10.1109/SP40001.2021.00076
4. Cao, Y., Xiao, C., Cyr, B., Zhou, Y., Park, W., Rampazzi, S., Chen, Q.A., Fu, K., Mao, Z.M.: Adversarial sensor attack on LiDAR-based perception in autonomous driving. In: ACM SIGSAC Conference on Computer and Communications Security (CCS). pp. 2267–2281 (2019). https://doi.org/10.1145/3319535.3339815
5. Cheng, Z., Choi, H., Liang, J.C., Feng, S., Tao, G., Liu, D., Zuzak, M., Zhang, X.: Fusion is not enough: Single-modal attacks on fusion models for 3D object detection. In: International Conference on Learning Representations (ICLR) (2024)
6. Huang, T., Liu, Z., Chen, X., Bai, X.: EPNet: Enhancing point features with image semantics for 3D object detection. In: European Conference on Computer Vision (ECCV). LNCS, vol. 12360, pp. 35–52. Springer (2020). https://doi.org/10.1007/978-3-030-58555-6_3
7. Ku, J., Mozifian, M., Lee, J., Harakeh, A., Waslander, S.L.: Joint 3D proposal generation and object detection from view aggregation. In: IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). pp. 1–8 (2018). https://doi.org/10.1109/IROS.2018.8594049
8. Liu, Z., Tang, H., Amini, A., Yang, X., Mao, H., Rus, D.L., Han, S.: BEVFusion: Multi-task multi-sensor fusion with unified bird's-eye view representation. In: IEEE International Conference on Robotics and Automation (ICRA). pp. 2774–2781 (2023). https://doi.org/10.1109/ICRA48891.2023.10160968
9. Madry, A., Makelov, A., Schmidt, L., Tsipras, D., Vladu, A.: Towards deep learning models resistant to adversarial attacks. In: International Conference on Learning Representations (ICLR) (2018)
10. Park, J., Park, K., Kim, D., Park, J., Choi, J.W.: Layer-wise modality decomposition for interpretable multimodal sensor fusion. In: Advances in Neural Information Processing Systems. vol. 38 (2025), arXiv:2511.00859
11. Sadeghian, R., et al.: Reliability-driven LiDAR-camera fusion for robust 3D object detection. arXiv preprint arXiv:2502.01856 (2025)

12. Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D.: Grad-CAM: Visual explanations from deep networks via gradient-based localization. In: IEEE International Conference on Computer Vision (ICCV). pp. 618–626 (2017). https://doi.org/10.1109/ICCV.2017.74
13. Thys, S., Van Ranst, W., Goedemé, T.: Fooling automated surveillance cameras: Adversarial patches to attack person detection. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW). pp. 49–55 (2019). https://doi.org/10.1109/CVPRW.2019.00012
14. Tu, J., Li, H., Yan, X., Ren, M., Chen, Y., Liang, M., Bitar, E., Yumer, E., Urtasun, R.: Exploring adversarial robustness of multi-sensor perception systems in self driving. In: Conference on Robot Learning (CoRL). PMLR, vol. 164, pp. 1013–1024 (2021)
15. Valada, A., Mohan, R., Burgard, W.: Self-supervised model adaptation for multimodal semantic segmentation. International Journal of Computer Vision **128**, 1239–1285 (2020). https://doi.org/10.1007/s11263-019-01188-y
16. Vora, S., Lang, A.H., Helou, B., Beijbom, O.: PointPainting: Sequential fusion for 3D object detection. In: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 4603–4611 (2020). https://doi.org/10.1109/CVPR42600.2020.00466
17. Yan, J., Liu, Y., Sun, J., Jia, F., Li, S., Wang, T., Zhang, X.: Cross modal transformer: Towards fast and robust 3D object detection. In: IEEE/CVF International Conference on Computer Vision (ICCV). pp. 18268–18278 (2023)
18. Yeong, D.J., Panduru, K., Walsh, J.: Exploring the unseen: A survey of multi-sensor fusion and the role of explainable AI (XAI) in autonomous vehicles. Sensors **25**(3), 856 (2025). https://doi.org/10.3390/s25030856
19. Zeng, Z., Yin, Y., Yuan, Z., Dey, A., Bao, X.: RESAR-BEV: An explainable progressive residual autoregressive approach for camera-radar fusion in bev segmentation. IEEE Transactions on Intelligent Transportation Systems (2026). https://doi.org/https://doi.org/10.1109/TITS.2026.3672182